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› Miuzei 20KG High Torque Waterproof Digital Servo Motor Instruction Manual

Miuzei 20kg servo

Miuzei 20KG High Torque Waterproof Digital Servo Motor Instruction Manual

Model: DS3218

Brand: Miuzei

1. INTRODUCTION

Thank you for choosing the Miuzei 20KG High Torque Waterproof Digital Servo Motor. This high-performance servo is engineered for precision and durability, featuring metal gears and a waterproof design suitable for a wide range of remote-controlled models, including 1/8, 1/10, and 1/12 scale R/C cars, robots, and airboats. This manual provides essential information for setup, operation, maintenance, and troubleshooting to ensure optimal performance and longevity of your servo.



Figure 1: Miuzei 20KG Digital Servo Motor and included accessories.

2. KEY FEATURES

- **270° Digital Servo:** Built-in brush motor and stable motherboard, precise angle without deviation.
- **Control System:** Change of pulse width. Pulse range 500usec - 2500usec for 0° - 270° (90°--1166usec, 180°--1833usec).
- **High Performance:** High quality aluminum gear equipped with double ball bearing, less noise, more stable torque. After the upgrade of RC Servo, two functions are added, which can be automatically protected with anti-burning and rotor blocking protection; 5S overload protection, 70 °C overheat protection.
- **Large Torque:** It reacts quickly and the maximum torque is up to 22.8kg·cm (316.63oz·in) at 6.8V.
- **Application Fields:** Suitable for airboats, 1:8 / 1:10 rc cars, robots, intelligent DIY, also rigid wings, helicopters, and other RC models.

Metal gears

Low noise
No angle deviation



Reinforced bearings

Stabilize to avoid damage



Stable chip

High performance and safer



Sealed housing

IP66 Water



Figure 2: Internal features including metal gears for low noise, reinforced bearings for stability, a stable chip for high performance, and a sealed housing for IP66 water resistance.

Automatic Protection

With Anti-Burning And Locked-Rotor Protection

5S Overload Protection

70°C Overheat Protection

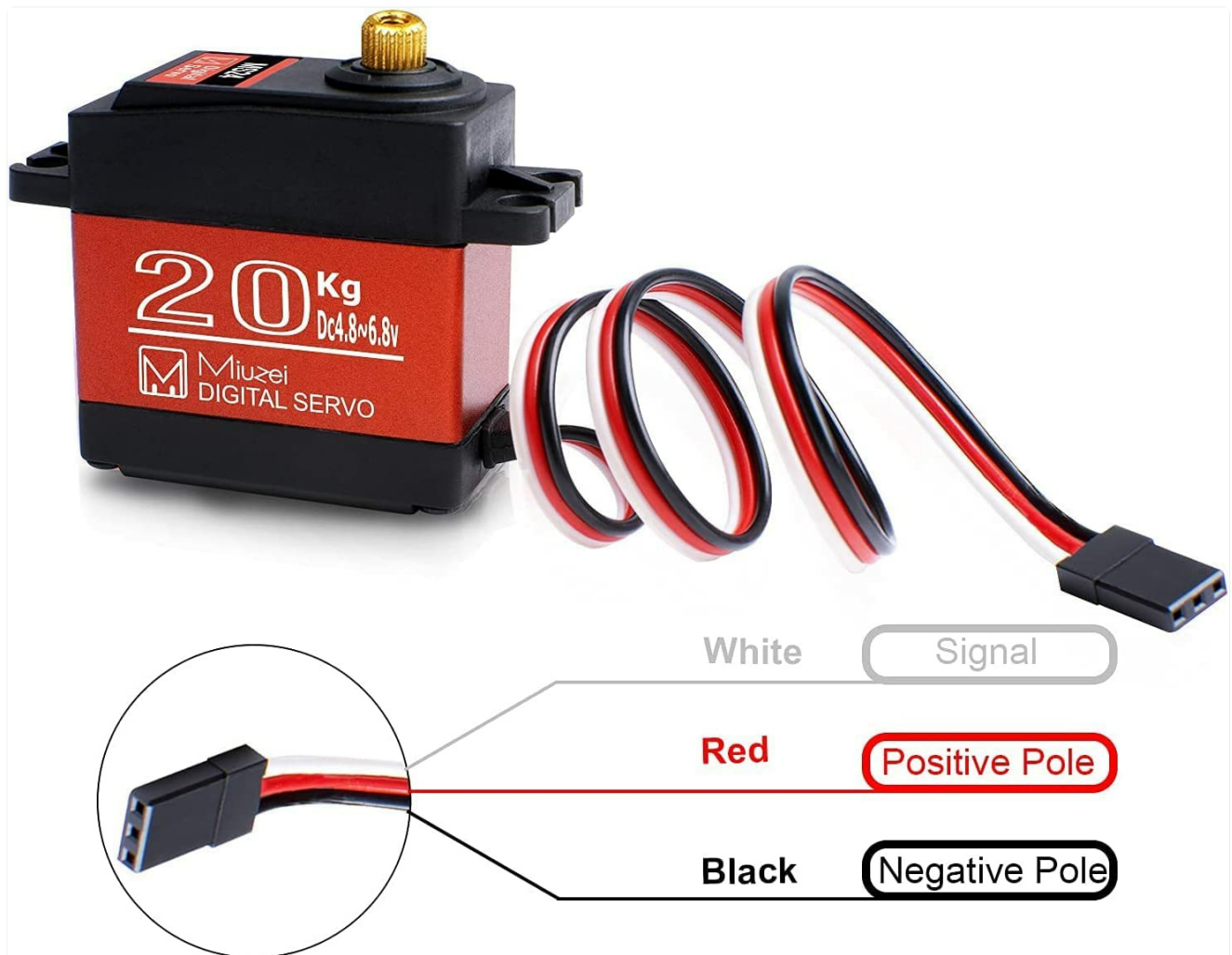


Figure 3: Automatic protection features for enhanced durability.

3. SETUP AND INSTALLATION

Proper connection is crucial for the servo's functionality and to prevent damage. Follow the wiring instructions carefully.

3.1 Wiring Diagram



WARNING !

please don't connect the wrong positive and negative poles when using it, otherwise the internal of the servo motor will short circuit.

Figure 4: Servo Wiring Diagram. Ensure correct polarity to avoid short circuits.

- **White Wire:** Signal (PWM input from receiver/controller)
- **Red Wire:** Positive Pole (Power supply, DC 4.8V-6.8V)
- **Black Wire:** Negative Pole (Ground)

WARNING: Please do not connect the wrong positive and negative poles when using it, otherwise the internal of the servo motor will short circuit.

3.2 Physical Installation

Mount the servo securely in your R/C model using the provided screws and mounting hardware. Ensure there is no obstruction to the servo arm's movement throughout its full range of motion. Select the appropriate servo horn (arm) from the included accessories based on your application's requirements.

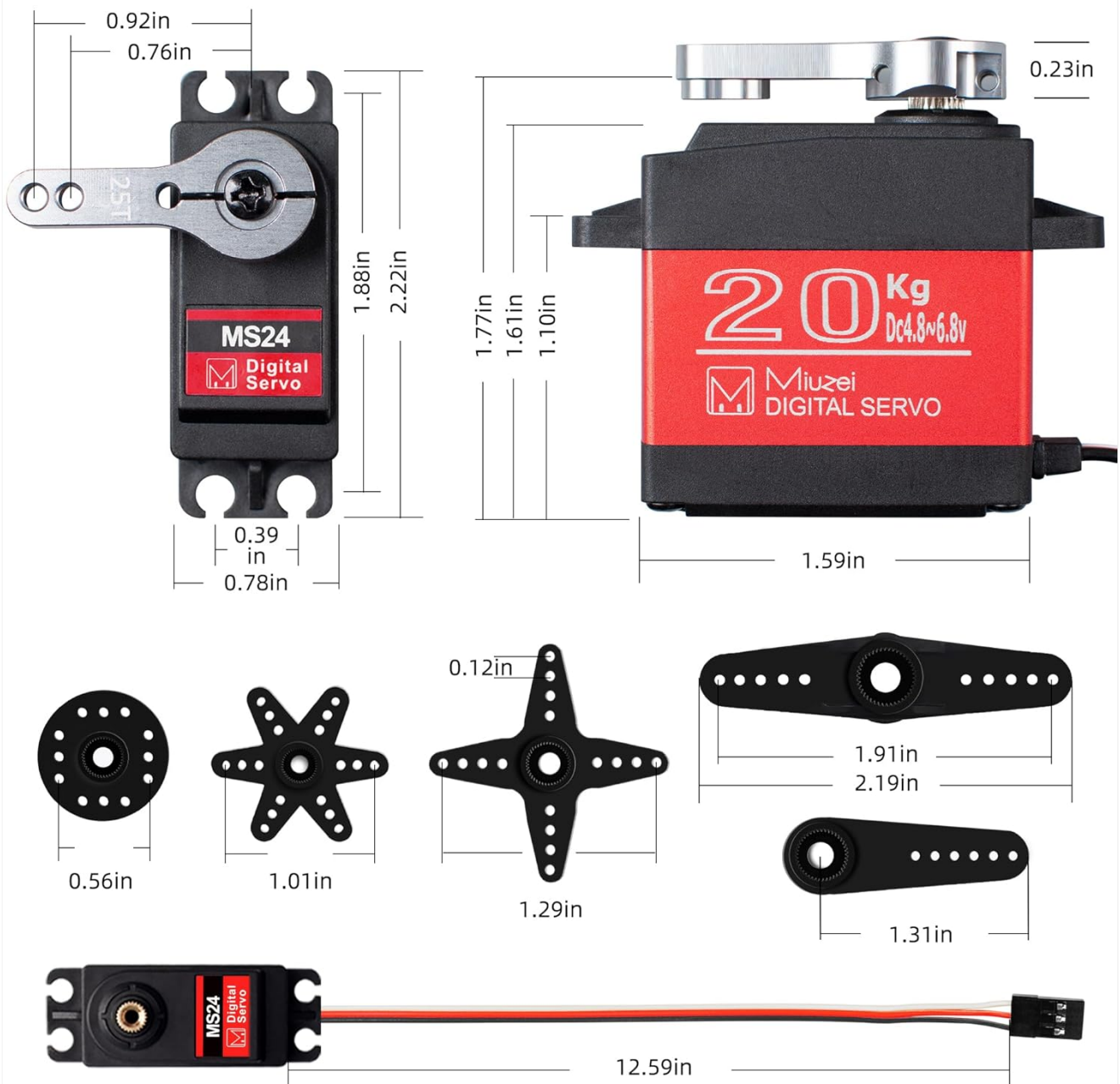


Figure 5: Servo dimensions and various servo horn types for different applications.

4. OPERATING INSTRUCTIONS

The Miuzei 20KG servo operates based on pulse width modulation (PWM) signals from your receiver or microcontroller. It offers a wide control angle of 0° to 270°.

4.1 Control Angle and Pulse Width

The servo's position is determined by the width of the input pulse:

- **Pulse Range:** 500usec - 2500usec for 0° - 270°
- **Neutral Position (90°):** Approximately 1166usec
- **180° Position:** Approximately 1833usec

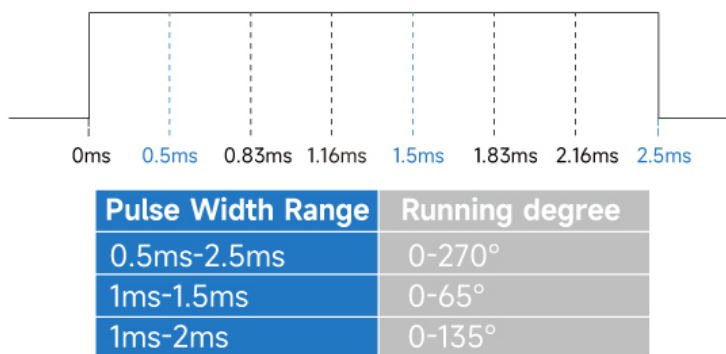


Figure 6: Pulse Width to Angle Mapping.

4.2 Performance Overview

The servo provides quick reaction times and stable torque, making it suitable for demanding applications. Its high-precision positioning system ensures accurate motion control.

Video 1: Demonstration of the Miuzei 20KG Servo Motor's operation and waterproof capabilities. This video showcases the servo's movement and its ability to function while submerged, highlighting its durability and precision.

5. MAINTENANCE AND CARE

To ensure the longevity and optimal performance of your Miuzei servo, follow these maintenance guidelines:

- **Waterproof Rating (IP66):** The servo is dust and splash water resistant. It prevents small solids from entering and large amounts of liquid splashes. This rating is suitable for indoor or semi-outdoor use.
- **Water Immersion:** When you use the servo in a water environment, generally 3 to 5 minutes of short-time immersion can work normally, but not for a long time immersion; the depth of water should not exceed 0.5 meters.
- **Cleaning:** Keep the servo clean from dirt, dust, and debris. Use a soft, dry cloth for cleaning. Avoid using harsh chemicals or solvents.
- **Storage:** Store the servo in a dry, cool place away from direct sunlight and extreme temperatures.
- **Inspection:** Periodically check the wiring for any signs of wear or damage. Ensure all connections are secure.



Waterproof



Fast Response



Strong Torque



Fast Heat
Dissipation



Low Noise



Limit Rotation Angle: 360°

Bandwidth: 3 μsec

Operating Speed:

0.18sek/60 Grad (4.8 V)

0.14sek/60 Grad (6.8V)

Size: 40.0 mm x 20.5 mm x 40.5 mm

Cable Length: 320mm $\pm$ 5m

Weight: 56G $\pm$ 5%

Operating Voltage: 4.8-6.8 V

Control Angle: 0° - 270°

Figure 7: Waterproof capability of the servo.

6. TROUBLESHOOTING

If you encounter issues with your Miuzei servo, consider the following common troubleshooting steps:

- **Servo Not Responding:**

- Check all wiring connections to ensure they are secure and correctly polarized (Signal, Positive, Negative).
- Verify the power supply voltage is within the specified range (4.8V-6.8V).
- Test with a different receiver or servo tester to rule out issues with the control signal source.

- **Erratic Movement or Jitter:**

- Ensure the power supply can provide sufficient current for the servo, especially under load.
- Check for electromagnetic interference (EMI) from other electronic components.
- Verify the control signal is stable and within the correct pulse width range.

- **Overheating:**

- Ensure the servo is not continuously stalled or overloaded.
- Check for mechanical binding in the system the servo is controlling.
- Verify adequate ventilation around the servo.

- **Limited Rotation Angle:**

- Confirm your control signal (e.g., from your transmitter or microcontroller code) is sending the full 500usec to 2500usec pulse width range to achieve the 270° rotation.
- Check for physical obstructions limiting the servo arm's movement.

If problems persist, please contact Miuzei customer support for further assistance.

7. PRODUCT SPECIFICATIONS

Attribute	Value
Brand	Miuzei
Model Name	rc servo
Item Model Number	20kg servo
Voltage	6.8 Volts (Operating Voltage: 4.8-6.8 V)
Horsepower / Torque	20 Kilograms (Max Torque: 22.8kg-cm at 6.8V)
Product Dimensions	1.59 x 1.59 x 1.77 inches; 2.89 ounces (40.0 mm x 20.5 mm x 40.5 mm)
Cable Length	320mm ±5m
Weight	56G ±5%
Control Angle	0° - 270°
Operating Speed	0.18sec/60 Grad (4.8V) \ 0.14sec/60 Grad (6.8V)
Bandwidth	3 usec
Manufacturer	Miuzei
Country of Origin	China

8. WARRANTY INFORMATION

Miuzei products are manufactured to high-quality standards. This product comes with a standard manufacturer's warranty against defects in materials and workmanship. Please retain your proof of purchase for warranty claims. For specific warranty terms and conditions, refer to the documentation included with your purchase or contact Miuzei customer support.

9. CUSTOMER SUPPORT

For any questions, technical assistance, or support regarding your Miuzei 20KG Servo Motor, please visit the official Miuzei store or contact our customer service team.

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